Return on Investment (ROI) Analysis

Dual-view financial model (internal cost vs. billable capacity), scenario sensitivity, Second Eye unit economics, and decision gates

Baseline Operational & Rate Assumptions

Parameter	Assumed Value	Validation
Monthly project volume	30 drawing sets / month	Replaced with measured volume in Week 1
Manual QC effort per set	12 hours / set (360 QC hrs/mo)	Timed across Architecture & MEP reviewers
Escaped issues & rework	2 issues/set × 3 hrs = 180 rework hrs/mo	Measured from historical RFIs & revision logs
Senior QC internal cost (View A)	$5.81 / hr (₹90,000/mo CTC ÷ 176 hrs)	True internal cost under fixed-fee AOP
Client billing rate (View B)	$22.00 / hr	Applies only when freed QC hours are sold
Pilot / Running cost	$2,769 one-time · $691 / month	From Costing model ($134 Cloud/LLM + $557 staff)

ROI Model Equations

Metric	Formula / Definition
Hours saved / month	(Sets × QC hrs × Time cut %) + (Sets × Escaped issues × Catch % × Rework hrs)
View A — Internal Cost Net	(Hours saved × $5.81/hr internal cost) − $691/mo running cost
View B — Capacity Net	(Hours saved × $22.00/hr billing rate) − $691/mo running cost (requires selling freed hours)
Payback & Year-1 ROI	Payback = $2,769 ÷ Monthly Net · Year-1 ROI = (12 × Monthly Net − $2,769) ÷ $2,769
Cost-View Break-Even	Requires ~20.5% (~21%) QC time reduction at base catch rate just to cover $691/mo

Three-Scenario Financial Comparison — View A (Internal Cost @ $5.81/hr) vs. View B (Billable Capacity @ $22/hr)

Scenario	QC Cut	Catch %	Hrs Saved/Mo	View A Gross	View A Net	View A Payback	View A Yr-1 ROI	View B Gross	View B Net	View B Payback	View B Yr-1 ROI
Conservative	15%	15%	54 + 27 = 81 hrs	$471	-$221 / mo	Does not pay back	-196%	$1,782	+$1,091 / mo	2.5 months	+373%
Base Case	25%	25%	90 + 45 = 135 hrs	$784	+$93 / mo	29.7 months	-60%	$2,970	+$2,279 / mo	1.2 months	+887%
Optimistic	35%	35%	126 + 63 = 189 hrs	$1,098	+$407 / mo	6.8 months	+76%	$4,158	+$3,467 / mo	0.8 months	+1402%

Second Eye Service Upside (Not in Headline ROI)

Unit Economics Item	Value	Notes
Assumed client price	$300 / set	Verify against current Second Eye pricing
Direct reviewer time (6 hrs @ $5.81) + LLM ($2)	$36.87 / set	AI-assisted coordination check
PM/QA (20%), Sales (15%) & Insurance (5%) allowances	$120.00 / set	40% total overhead & liability cover
Total delivery & overhead cost	$156.87 / set	Fully loaded cost per set
Net contribution margin per set	$143.13 / set (48%)	+$4,294 Year-1 contribution at 30 sets/yr

Executive Go / No-Go Decision Rule (End of Week 4)

Required Condition	Commercial Rationale
1. Measured QC time cut ≥ 30%	Exceeds the 21% internal-cost break-even with safety margin
2. Reviewer acceptance ≥ 85%	Ensures engineers save time rather than chasing false alarms
3. Covered issue recall ≥ 80%	Verified against 3–5 historical Uppteam project QC logs
4. Capacity redeployment plan	Management commits freed QC hours to additional AOP sets or Second Eye

The honest cost picture (View A)

At India QC salary rates ($5.81/hr), a 25% time cut saves $784/mo vs $691/mo cost (+$93/mo net, 29.7-mo payback). Wage savings alone do not justify the project.

Where the real ROI lives (View B)

When freed senior QC hours handle more fixed-fee AOP sets or billable work ($22/hr), the Base Case nets +$2,279/mo and pays back in 1.2 months (887% Yr-1 ROI).

Unquantified client-trust upside

Fewer escaped dimensional, MEP, and spec errors reaching U.S. clients avoids costly field RFIs, rework, and account churn.

Second Eye margin expansion

Cutting third-party QC review to ~6 hrs/set yields a 48% contribution margin ($143/set), adding $4,294/yr per 30 sets sold.